Pricing and Production without the Invisible Hand

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Presentation Plan

Motivations: Addressing the Arrow Critique

- **The Theoretical Gap:** The Arrow (1959) critique argues that standard perfectly competitive models lack a logical place for rational price adjustment.
- **The Modern Inversion:** Standard New Keynesian frameworks allow firms to set prices but force them to produce the demanded quantity, violating the principle of voluntary exchange.
- **Proposed Solution:** The paper introduces a macroeconomic framework where firms explicitly and simultaneously formulate rational decisions regarding both prices and production quantities under uncertainty.
- **Market Discrepancies:** Because these decisions are made prior to the realization of demand and productivity shocks, individual product markets can be in excess supply or rationed.

Decision-Making Under Uncertainty

- Firms must determine both price and labor inputs prior to the realization of idiosyncratic demand and productivity shocks.
- **Axioms of Exchange:** Exchange is strictly voluntary (nobody can be forced to trade more than they are willing) and efficient (no mutually beneficial trades fail to occur).
- **Market States:** This environment yields two distinct market states:
  1. **Rationed Market:** Demand exceeds supply. Marginal production is highly beneficial as the firm sells all available supply.
  2. **Slack Market:** Supply exceeds demand. Firms incur fixed labor costs for surplus production that yields no revenue.

The Firm's Problem

$$\sup_{p,L} \int_{\mathbb{R}_4^{++}} \Lambda(pq - wL)\,\mathrm{d}G(\Lambda, A, z, w)$$
$$\text{s.t. } q = \min\{q^s, q^d\},\ q^s = AL,\ q^d = zp^{-\eta}$$

- **The Kinked Objective:** The transacted quantity q is the minimum of supply and demand, which creates a non-standard kink in the firm's revenue function.
- **Firm Tightness (t):** $t \equiv \frac{p^{-\eta}}{L}$. This represents the firm's chosen balance of price-implied demand relative to labor.
- **Supply-Demand Shock (U):** $U \equiv \frac{A}{z}$. This represents the uncontrollable ratio of productivity to the demand shifter.
- **The Rationing Condition:** The market is rationed iff $t \geq U$.

The Firm's Problem

$$
\underbrace{(\eta-1)\mathbb{E}[\Lambda z p^{-\eta}|t \leq U]}_{\text{Revenue losses when slack}} \times \underbrace{(1-\mathbb{P}[t \geq U])}_{\text{Slack probability}}
$$

$$
= \underbrace{\mathbb{E}[\Lambda A L|t \geq U]}_{\text{Revenue gains when rationed}} \times \underbrace{\mathbb{P}[t \geq U]}_{\text{Rationing probability}}
$$

- **Balancing Marginal Effects:** At the optimal price, the expected losses from raising prices in a slack market must exactly equal the expected gains from raising prices in a rationed market, weighted by their respective probabilities.
- **Slack vs. Rationed Returns:** When the market is slack, a price increase causes revenue losses proportional to the demand elasticity. When rationed, the firm gains additional revenue on all produced units.

Markups

$$p^* = \frac{1}{r}\frac{\mathbb{E}[\Lambda w]}{\mathbb{E}[\Lambda A | t^* \geq U]}$$

$$\text{where: } r = P[t \geq U]$$

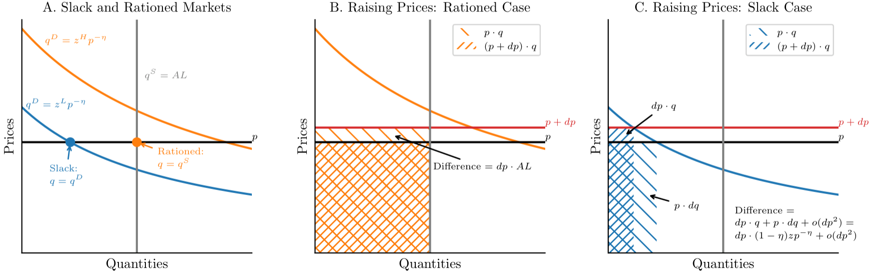
- **Uncertainty-Based Markups:** Optimal pricing corresponds to placing a markup of the inverse rationing probability on expected marginal costs.
- **Correcting for Inference:** This expectation also accounts for the inference that when a market is rationed, productivity is likely low and marginal costs are likely high.

The Revenue Kink and Microeconomic Behavior

- **The Revenue Kink:** The firm's profit function exhibits a structural kink exactly where supply equals demand, fundamentally altering pricing incentives.
- **Departure from the Lerner Formula:** Optimal monopoly pricing is decoupled from the classical Lerner formula; markups are optimally determined by the inverse probability of market rationing.
- **Microeconomic Contraction:** Volatility imposes first-order effects on firm behavior. In response to heightened uncertainty, firms optimally "shrink" by raising prices and reducing labor inputs to self-insure against the downside risk of slack markets.

Decision-Making Under Uncertainty

Figure 1: The Incentives to Raise Prices



General Equilibrium: The Household Problem

- **Household Preferences:** Households have constant elasticity of substitution preferences over a continuum of consumption varieties, indexed by $i \in [0, 1]$.

$$C = \left(\int \theta_i^{\frac{1}{\eta}} c_i^{\frac{\eta-1}{\eta}} di \right)^{\frac{\eta}{\eta-1}}$$

- **Utility Structure:** Households possess Golosov and Lucas preferences over the consumption aggregate, money, and labor supply.

$$\ln C + \ln \left(\frac{M}{P} \right) - \alpha N$$

- **Consumption Constraints:** A key departure from standard models is that households cannot consume more of any specific good than the amount that has actually been produced ($\overline{c}_i$).

$$c_i \leq \overline{c}_i \quad \text{for } i \in [0, 1]$$

General Equilibrium: Equilibrium Aggregates

- **Equilibrium Aggregates:** By combining the household's optimal behavior with the firm's optimality conditions, the model endogenously determines equilibrium mean demand (μ_z) in non-rationed markets.

$$\mu_z = \ln\left[\left(\frac{\eta}{\eta-1}\frac{\alpha}{e^{\mu_A}}\right)^{\eta-1}\overline{M}^{\eta}\right] + \underbrace{\eta\phi(\kappa)\sigma_U}_{\text{Uncertainty Spillover}}$$

General Equilibrium: First-Order Effects

- **First-Order Effects:** In the general equilibrium setting, an increase in uncertainty (σ_U) raises firms' prices (p) and reduces aggregate consumption (C).

$$\ln p = \ln\left(\frac{\eta}{\eta-1}\frac{\alpha M}{e^{\mu_A}}\right) + \left(\phi(\kappa) - \frac{\kappa}{\eta}\right)\sigma_U$$

$$\ln L = \ln\left(\frac{\eta-1}{\eta}\frac{1}{\alpha}\right)$$

$$\ln C = \ln\left(\frac{\eta-1}{\eta}\frac{e^{\mu_A+\frac{\mu_\theta}{\eta-1}}}{\alpha}\right) - \left(\phi(\kappa) - \frac{\kappa}{\eta}\right)\sigma_U$$

General Equilibrium: Constraints and Spillovers

- **The Household Constraint:** In general equilibrium, consumers are physically constrained; they cannot consume a higher quantity of any specific variety than what has been produced.
- **Cross-Market Substitution:** When consumers encounter stockouts (rationed markets), they substitute their unspent expenditures toward other available varieties.
- **Negative Aggregate Demand Externality:** Consequently, an expansion of production in rationed markets directly diminishes the spillover demand allocated to non-rationed (slack) markets.

General Equilibrium Outcomes

- **The Labor Offset:** Surprisingly, total labor inputs do not respond to uncertainty shocks to the first order. The firm's partial equilibrium desire to reduce labor is perfectly neutralized by general equilibrium demand spillovers.
- **Uncertainty Shocks:** Increases in uncertainty function analogously to markup shocks, yielding first-order effects that uniformly raise aggregate prices and depress aggregate consumption.
- **Monetary Policy Transmission:** Unanticipated demand shocks boost aggregate consumption mechanically by reducing economic slack, allowing households to purchase previously overproduced goods.

The New Old Keynesian (NOK) Model

- **Dynamic Framework:** The static model is extended into an infinite-horizon setting utilizing Calvo (1983) pricing and input stickiness, where a fixed fraction of firms cannot adjust prices or production quantities in a given period.
- **Key Theoretical Departures from the Standard NK Benchmark:**
  1. **Uncertainty Shocks:** Increases in idiosyncratic risk act as endogenous cost-push shocks, generating output contractions alongside elevated inflation.
  2. **Procyclical Productivity:** Demand expansions absorb existing product-market slack, raising measured labor productivity without requisite shifts in physical technology.
  3. **Transmission of Demand Shocks:** Short-run consumption responses are driven primarily by changes in market tightness rather than instantaneous labor adjustments.

The Linearized NOK Model: A Three-Equation System

- **1. The New Keynesian Phillips Curve (NKPC):**

$$\pi_\tau = \Gamma(\hat{M}_\tau - \overline{p}_\tau) + \beta \mathbb{E}_\tau[\pi_{\tau+1}] + \zeta \hat{\sigma}_{U,\tau}$$

Implication: First-order conditional price dynamics coincide with the standard NK model; expected future rationing does not alter the linearized Phillips curve.

- **2. The NOK Tightness Curve:**

$$\upsilon_\tau = -\Gamma(\hat{D}_\tau + \overline{t}_\tau) + \beta \mathbb{E}_\tau[\upsilon_{\tau+1}] + \psi \hat{\sigma}_{U,\tau}$$

- **3. The Demand Spillovers Curve:**

$$\hat{D}_\tau = \gamma_M \hat{M}_\tau + \gamma_p \overline{p}_\tau + \gamma_t \overline{t}_\tau + \gamma_U \hat{\sigma}_{U,\tau}$$

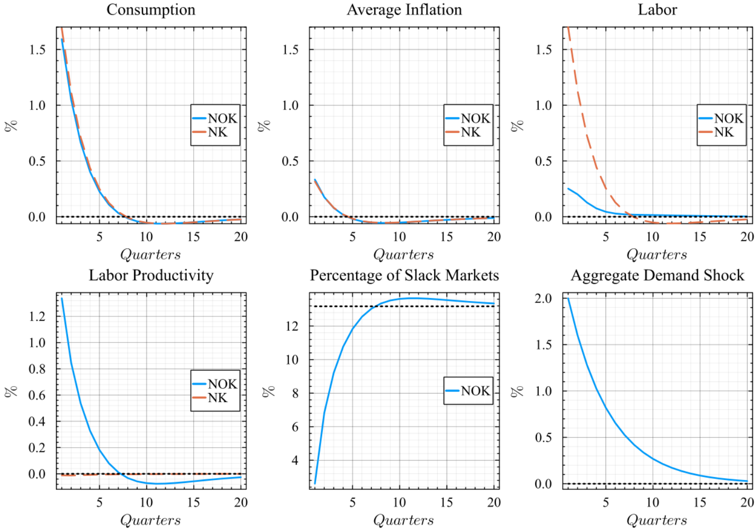
Implication: Real shock transmission relies fundamentally on the dynamic coevolution of aggregate market tightness and demand spillovers.

Dynamics: Monetary Policy Transmission

- **Inflation and Consumption Dynamics:** The aggregate paths approximate the standard NK benchmark due to the equivalence of the NKPC to a first-order approximation.
- **The Employment Margin:** Unlike the standard NK model, which assumes the Walrasian auctioneer necessitates immediate labor expansion to clear markets, the NOK framework accommodates demand shocks primarily through the absorption of existing product-market slack.
- **Measured Productivity:** The disproportionate response of transaction volumes relative to labor inputs yields a mechanical, short-run increase in measured labor productivity.

Dynamics: Monetary Shocks (Impulse Responses)

Figure 4: Impulse Responses to an Aggregate Demand Shock

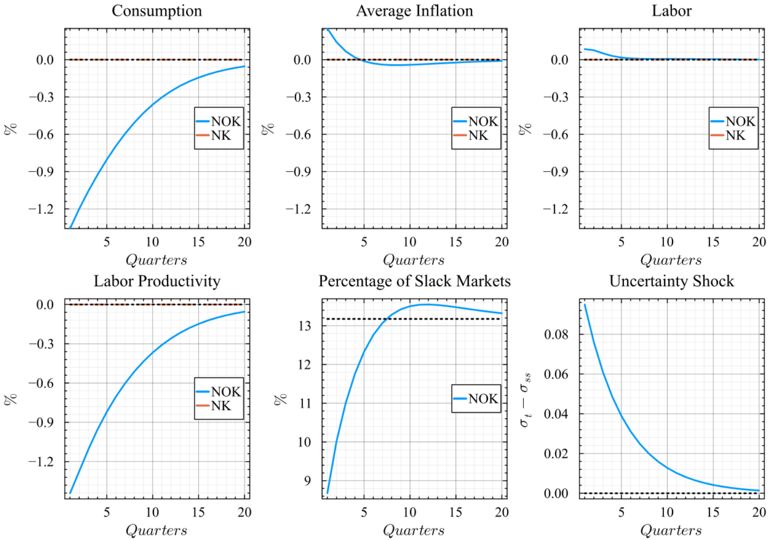


Dynamics: Uncertainty Shock Transmission

- **The Walrasian Benchmark:** Under the standard NK model, idiosyncratic uncertainty does not alter aggregate allocations to a first-order approximation, as continuous market-clearing is assumed.
- **Firm-Level Response:** In the NOK framework, firms optimally self-insure against adverse demand realizations by elevating prices and reducing labor inputs.
- **Aggregate Implications:** Elevated prices reduce transaction volumes and induce stock-outs. This manifests structurally as a decline in measured aggregate productivity (the Solow residual), holding underlying physical technology constant.

Dynamics: Uncertainty Shocks (Impulse Responses)

Figure 5: Impulse Responses to an Uncertainty Shock



Concluding Remarks

- **Theoretical Framework:** The paper develops a general equilibrium model wherein prices and production quantities are determined jointly by firms under uncertainty, relaxing the standard New Keynesian reliance on a Walrasian auctioneer to clear markets.
- **Static Equilibrium Properties:**
  - *Pricing:* Optimal monopoly pricing diverges from the Lerner (1934); markups are determined by the inverse probability of market rationing.
  - *First-Order Uncertainty Effects:* Idiosyncratic uncertainty acts as a first-order friction, elevating prices and depressing labor demand.
  - *Spillovers:* Production in rationed markets generates negative aggregate demand externalities for slack markets.
- **Dynamic Implications (NOK Model):**
  - When augmented with Calvo (1983) frictions, the model provides an endogenous mechanism for stagflationary uncertainty shocks.
  - The framework aligns with recent empirical evidence regarding the procyclicality of measured labor productivity and the differential response lags of sales versus employment to monetary shocks.